Name _____ Period _____

Solving Problems | STUDENT EXEMPLAR: ONE POSSIBLE SOLUTION

Your Tasks (Mark these off as you go)

- ☐ Find the file using the most storage
- ☐ Sort the files from largest to smallest
- ☐ Implement a quicksort
- ☐ Compare algorithms
- ☐ Receive credit for this lab guide

☐ Find the file using the most storage

Your device is almost out of storage. Eight files use different amounts of storage. Choose one group member to serve as the computer and hold the File Size Key. How could the rest of the group identify the file using the most storage? Follow these rules:

- Only the computer may look at the File Size Key.
- The group may ask the computer to compare only two files at a time. The computer reports only which file is larger; no file sizes may be revealed.
- You may rearrange the file cards and tally comparisons, but you may not record file sizes or comparison results.

Write an algorithm below that another group could implement.

EXEMPLAR RESPONSE

Choose a current largest file. Start with any one of the eight cards as the current largest.

Compare. Ask the computer to compare the current largest with one file that has not yet been checked.

Keep the winner. The larger of the two becomes the new current largest. Move the smaller file aside.

Repeat. Continue until every other file has been compared. The remaining current largest is the file using the most storage.

Why it works. Each comparison eliminates one file from being the largest, so only one file remains.

Obtain eight file cards and a File Size Key from your teacher. Choose the computer and implement your algorithm to identify the file using the most storage. Keep track of the number of comparisons and record it below.

EXEMPLAR RESPONSE

Comparisons: 7

Largest file: File C (the computer does not reveal its size to the group).

□ Sort the files from largest to smallest

Brainstorm an algorithm you could use to sort the eight files from largest to smallest. Continue to follow the same roles and comparison rules:

- Only the computer may look at the File Size Key.
- Only two files may be compared at a time. The computer reports only which file is larger.
- You may rearrange the file cards and tally comparisons, but you may not record file sizes or comparison results.

Write an algorithm below that another group could implement.

EXEMPLAR RESPONSE

First position. Use the previous algorithm to find the largest file among all eight cards. Place it in the first position of the sorted row.

Next positions. Find the largest file among the cards that remain and place it in the next open position.

Finish. Repeat until only one card remains. It belongs in the last position. This strategy is selection sort because each pass selects the largest remaining item.

Implement your algorithm to sort the files from largest to smallest. Keep track of the number of comparisons and record it below.

EXEMPLAR RESPONSE

Comparison count: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28

Final order: File C, File F, File A, File H, File D, File G, File B, File E

Check: Each pass places the largest remaining file, so the completed row runs from largest to smallest.

Using your algorithm, predict how many comparisons it would take to sort 10 files, 20 files, and 100 files. Explain how you made each prediction.

EXEMPLAR RESPONSE

Pattern: Selection sort uses (n - 1) + (n - 2) + ... + 1 comparisons, which is n(n - 1) / 2.

10 files: 10(9) / 2 = 45 comparisons

20 files: 20(19) / 2 = 190 comparisons

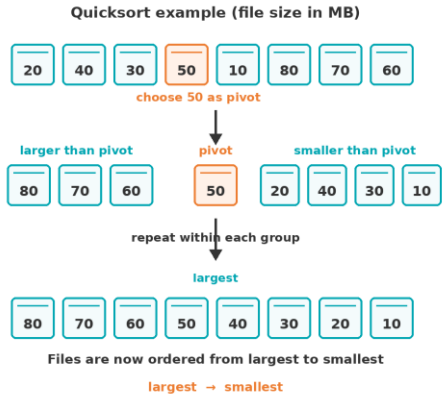
100 files: 100(99) / 2 = 4,950 comparisons

□ Implement a quicksort

Quicksort is a comparison-based sorting algorithm that can be much more efficient than a simple sorting method, particularly for large collections. Its efficiency depends on the pivot selected.

Below is the quicksort algorithm as it applies to the file cards:

- Choose one file at random as the pivot and place it aside.
- Compare each remaining file with the pivot. Put files that are larger on the left and files that are smaller on the right. One side may contain more files than the other.
- Choose one group and repeat this procedure. Then do the same for the other group. Keep each pivot between its larger and smaller groups.
- Continue partitioning the remaining groups until no group contains more than one file.
- When every group has been reduced to a single file, the files will be ordered from largest to smallest.



Mix up your file cards and apply the quicksort algorithm above. The computer should continue to reveal only which of two files is larger. Keep track of the number of comparisons and record it below.

EXEMPLAR RESPONSE

Pivot File D: 7 comparisons -> larger [A, C, F, H] | D | smaller [B, E, G]

Larger group, pivot File F: 3 comparisons -> [C] | F | [A, H]. Then pivot A: 1 comparison -> [A, H].

Smaller group, pivot File B: 2 comparisons -> [G] | B | [E].

Result: C, F, A, H, D, G, B, E. Total = 7 + 3 + 1 + 2 = 13 comparisons.

□ Compare algorithms

How did the number of comparisons used by your original algorithm differ from the number used by quicksort?

EXEMPLAR RESPONSE

Selection sort used 28 comparisons. Quicksort used 13. For this pivot sequence, quicksort used 15 fewer comparisons. Selection sort keeps rescanning most of the remaining cards, while quicksort divides the problem into smaller groups.

If each pivot divided the remaining files as evenly as possible, how many comparisons would quicksort use to order all eight files? Show or explain your reasoning.

EXEMPLAR RESPONSE

First pivot: 7 comparisons divide the other seven files into groups of 3 and 4.

Group of 3: 2 comparisons finish this balanced group.

Group of 4: 3 comparisons split the group, then 1 comparison finishes the remaining pair.

Total: 7 + 2 + 3 + 1 = 13 comparisons.

If you always chose the largest or smallest remaining file as the pivot, how many comparisons would quicksort use to order all eight files? Show or explain your reasoning.

EXEMPLAR RESPONSE

Each pivot would leave every other file on the same side. The group sizes would shrink only from 8 to 7 to 6 and so on.

Total: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28 comparisons. This is quicksort's worst case for eight files.

□ Receive credit for this lab guide

Submit this portion of the lab to Pluska to receive credit for the lab guide.